The Psychological Record
https://doi.org/10.1007/s40732-021-00457-y

ORIGINAL ARTICLE

Assessing the Development of Relational Framing in Young Children

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Accepted: 1 February 2021

Abstract

Relational frame theory (RFT) sees operant acquisition of various patterns of relational framing (frames) as key to linguistic and cognitive development, and it has explored the emergence of a range of psychological phenomena (e.g., analogy, perspective-taking) in these terms. One potentially important advance for RFT research is to obtain more detailed information on the normative development of relational framing in childhood. This was one of the aims of the present study, which sought to measure relational responding of various types and at various levels of complexity in young children across a range of ages. A second aim of the study was to focus in particular on analogy, or the relating of relations, as one particularly important pattern of relational responding. The present study examined a range of frames including coordination, comparison, opposition, temporality, and hierarchy at a number of different levels of complexity (nonarbitrary relating, nonarbitrary relating of relations, arbitrarily applicable relating, and arbitrarily applicable relating of relations) in young children ranging in age from 3 to 7. Performance overall as well as under various subheadings was correlated with both age and intellectual ability. Outcomes and their implications are discussed.

Keywords Relational frame theory · Relational assessment · Normative development · Analogical relations · Children · Derived relations

Relational frame theory (RFT) is a contextual behavioral account of human language and cognition (Hayes et al., 2001a; Stewart, 2016; Stewart & Roche, 2013) that views arbitrarily applicable derived relational responding (AADRR), or relational framing, as the key operant underlying these repertoires. Many species can be trained to engage in nonarbitrarily applicable relational responding (NAARR), which involves relating stimuli based on their physical properties (e.g., selecting a stimulus physically similar, different, smaller, or larger than another). However, humans alone can learn AADRR, which involves relating stimuli based on contextual cues that specify the relation rather than on the formal or physical properties of those stimuli.

For example, consider AADRR of comparison. If I show a sufficiently verbally advanced child two novel, similarly sized coins, A and B, and tell them that A is worth less than B, then they will readily derive that B is worth more than A. This derivation is based on the cues "more" and "less" rather than on physical properties and, from an RFT perspective, is the result of exposure to multiple exemplar training (MET) provided by the socioverbal community in which this particular pattern of comparison framing has been reinforced across a variety of different stimuli in the presence of these cues. In this and many other frames, such training initially involves nonarbitrary relations (e.g., learning to pick a relatively smaller or larger quantity of items on being asked to select "less" or "more," respectively) but eventually the child can apply the response pattern in purely arbitrary situations in which nonarbitrary relations (e.g., physical size) no longer function as a guide, such as in the coin example.

According to RFT, all forms of framing are characterized by three properties, and it is these properties that underlie the substantial generativity of language (Stewart et al., 2013). Mutual entailment is the property of bidirectionality of stimulus relations, whereby if A is related to B, then B will be related to A (e.g., if A is less than B, then B is more than A). Combinatorial entailment involves the combination of previously acquired relations to allow derivation of novel relations (e.g., if A is less than B, and B is less than C, then A is less than C, and C is more than A). Transformation of functions is the property whereby the psychological functions of a

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Published online: 15 April 2021

stimulus in a derived relation can change depending on the functions of other stimuli in the relation as well as the nature of the relation itself. For example, if a child has already learned that A has monetary value and then derives that a novel coin C is worth more than A, they will choose C over A despite the fact that C is novel. In an alternative scenario, if an arbitrary stimulus A has acquired an aversive function (e.g., by being paired with shock), then despite not being directly paired with shock itself, C might come to be perceived as even more aversive than A based on the derived "larger than" relation. Indeed, such an outcome has been empirically demonstrated (see, e.g., Dougher et al., 2007).

RFT research has by now provided evidence of a variety of relational frames. Coordinate (sameness) framing or equivalence, as initially demonstrated by Sidman (1971), is the earliest and most researched example of AADRR, but subsequent research has provided evidence of various other patterns of relational framing, including difference (e.g., Steele & Hayes, 1991), opposition (Barnes-Holmes et al., 2004), comparison (Dymond & Barnes, 1995), hierarchy (Gil et al., 2014), analogy (Barnes et al., 1997), temporality (O'Hora et al., 2005), and deixis (McHugh et al., 2004).

Furthermore, in accordance with the thesis that relational framing is an operant, studies have not simply demonstrated relational framing but have successfully assessed and trained various patterns of framing in young children. For example, Barnes-Holmes, Barnes-Holmes, and Smeets (2004) successfully trained comparative AADRR as generalized operant behavior using multiple exemplar training (MET) in 4- to 6-year old children and this outcome has since been successfully replicated and extended (Berens & Hayes, 2007; Gorham et al., 2009). In several of these studies, some participants who did not readily learn to derive the arbitrary relations were aided in doing so via additional training in nonarbitrary relations, which emphasized the importance of NAARR as a prerequisite repertoire for AADRR. Apart from demonstrating that frames not already in place can be established, RFT research has also shown that already acquired framing repertoires can be substantially strengthened and that doing so can affect general cognitive ability (Cassidy et al., 2011; Hayes & Stewart, 2016; Mulhern et al., 2018).

Hence, RFT has provided substantive evidence that AADRR underlies human language and cognition (Hayes et al., 2001a), including work showing that training this repertoire can boost intellectual skill. However, despite the evidence that relational framing is a core skill underlying cognitive ability, research on the sequence of frame acquisition and the normative development of relational responding remains limited. Luciano et al. (2009) proposed a training sequence for early frames of coordination, opposition, distinction, comparison, and hierarchy and provided suggestions for intervention, including multiple exemplar training, bidirectional relational training, and transitioning from nonarbitrary to arbitrary relations. However, as yet there is little or no empirical work on the normative acquisition of frames. Previous research on relational framing and age has found that particular relational repertoires (deictic and hierarchical, respectively) increase as a function of age in typically developing children (McHugh et al., 2004; Mulhern et al., 2017). There is tentative evidence suggesting that equivalence emerges at a young age (Lipkens et al., 1993), but on a number of other basic frames there is little relevant research. Barnes-Holmes, Barnes-Holmes, and Smeets (2004) trained comparison in 4- to 6-year-old children, and Barnes-Holmes, Barnes-Holmes, and Smeets (2004) trained opposition in 4- to 6-year-old children, but no evidence was provided to suggest when these frames tend to emerge under natural conditions. In the case of other frames such as distinction and temporality, for example, there is as yet no relevant research. A key aim of the current cross-sectional study therefore was to provide more comprehensive data on the emergence of basic patterns of framing across ages 3 to 7 years. In this study, we investigated the normative development of a number of specific relational frames (coordination, comparison, opposition, temporality, and hierarchy) in young children and measured relational framing against standardized tests of cognitive abilities. This is one of the first attempts to provide a cross-sectional investigation of the acquisition of a variety of frames in a number of different age groups of children.

Apart from this, another focus of this relational assessment and of the study in general was on the relating of relations, or analogical responding, an important form of relational framing, and one that is relevant for intellectual development (Gentner & Christie, 2010; Goswami & Brown, 1989; Sternberg, 1977). Analogy is pervasive in human language and critically important in various key domains of human activity, including science, technology, and education. Furthermore, it is frequently used as a metric of intelligent behavior (e.g., the Miller Analogies Test, the Law School Admissions Test; Gentner, 1983; Morsanyi & Holyoak, 2010; Sternberg, 1977; Stewart et al., 2004, 2020). Given the apparent importance of analogy for intellectual development, cognitive developmental psychologists have examined the emergence of this skill in young children. Early researchers believed that analogical reasoning developed at the age of 12 or later, and that children younger than this relied on simple associative strategies (Levinson & Carpenter, 1974; Lunzer, 1965; Piaget et al., 1977/Piaget et al., 2001; Sternberg & Nigro, 1980). However, it has been argued that children as young as four can show analogical reasoning (Goswami & Brown, 1990) with prior knowledge playing a critical role. For example, Goswami and Brown (1989) examined the effect of children's previous experiences and found that children as young as 3 could complete analogical tasks when they had relevant knowledge about the domains involved (see also Alexander et al., 1989; Goswami, 1989).

Although research on analogy has been mostly the province of cognitive psychologists, during the last 2 decades behavior analysts have also begun to research analogy. The impetus for this has primarily come from researchers who take an RFT perspective.

Working within an RFT framework, Barnes et al. (1997) provided the first functional analytic definition of analogy as the derivation of a sameness or equivalence relation between equivalence relations, called "equivalence–equivalence" responding. For example, consider the analogy *apple* is to *orange* as *dog* is to *sheep*. In this case, *apple* and *orange* participate in an equivalence relation in the context of fruit; and *dog* and *sheep* participate in an equivalence relation in the context of animal, and thus, because these are both equivalence relations, we can derive a relation of equivalence between the relations themselves. In order to empirically model this phenomenon, Barnes et al. first trained and tested four 3-member equivalence relations in adults and 9-year-old children. They used a matching-to-sample (MTS) procedure to train conditional discriminations amongst three-letter nonsense syllables (coded using alphanumeric designations) as follows: A1→B1, A1→C1, A2→B2, A2→C2, A3→B3, A3→C3, A4→B4, A4→C4, and then tested for derivation of the following four untrained relations: B1↔C1, B2↔C2, B3↔C3, B4↔C4. After participants passed these equivalence tests, they were then tested for the derivation of equivalence relations between equivalence (and nonequivalence) relations themselves (i.e., equivalence–equivalence responding). This involved using compound stimuli comprised of either two nonsense syllables that were equivalent or two that were nonequivalent. Participants were required to choose an equivalent pair in the presence of an equivalent pair (i.e., equivalence–equivalence) and a nonequivalent pair in the presence of a non-equivalent pair (i.e., nonequivalence–nonequivalence). For example, given B3C3 and B3C4 as comparisons, if the sample was B1C1 then they had to choose B3C3, whereas if B1C2 was the sample, they had to choose B3C4. All participants related equivalence relations to other equivalence relations, and nonequivalence relations to other nonequivalence relations, and thus this constituted a basic model of analogical reasoning.

In the wake of Barnes et al. (1997), one stream of research used their equivalence-equivalence model to investigate the emergence of analogy in young children (Carpentier et al., 2002, 2003). Carpentier et al. (2002) used the equivalence–equivalence paradigm to investigate analogy in a range of age groups including adults and 9- and 5-year-old children. As in the original Barnes et al. study, they found that adults and 9-year-old participants readily showed equivalence–equivalence responding. In contrast, the 5-year-old children, although readily passing equivalence testing, initially failed to show equivalence–equivalence responding and required additional training before doing so. In particular, they required training and testing with compound–compound matching tasks with trained relations (e.g., A1B1–A3B3 and A1B2–A1B3) before they could successfully pass the derived compound relations (BC–BC) test. Carpentier et al. (2003) extended this work by assessing whether this additional training could also facilitate the 5-year-old children's ability to pass equivalence–equivalence tests before receiving the prior equivalence tests. This was something that Barnes et al. had shown that adults and 9-year-old children could do and this was replicated by Carpentier et al. (2003). However, despite providing considerable additional training, only 2 of 18 of the 5-year-old participants were successful in this task. The Carpentier et al. (2002, 2003) studies thus provided additional insight into the development of equivalence–equivalence responding as a functional analytic model of analogy. By providing a precise, functional-analytic model of this behavior it could be argued that this work has shed additional light on this phenomenon beyond that provided by mainstream, cognitive psychological work by not only confirming a developmental divide in the analogical ability at a particular age but also suggesting how additional training might remediate in this respect.

In summary, analogy is an important type of emergent intellectual ability. RFT has examined and modeled analogy in young children and provided an important contribution to the understanding of the acquisition of this ability. Thus, RFT has offered a substantive model of analogy, and has identified analogy as an important emergent repertoire. Hence, we wanted to focus on this particular ability, and placed an emphasis on analogy in our relational assessment protocol. One way in which we wished to extend the previous work was by also investigating nonarbitrary analogy, or relational matching, which RFT research has not examined before. We included this in addition to arbitrary analogy because as suggested above in the introduction, nonarbitrary relational responding is an important foundation for arbitrary relational responding.

In the present study, we looked at relational framing across a number of different frames (including coordination, comparison, opposition, temporality, and hierarchy), across four different levels (including nonarbitrary relations, nonarbitrary analogy, arbitrary relations, and arbitrary analogy), and across a number of different age groups (in particular, 3- to 4-, 4- to 5-, 5- to 6-, and 6- to 7-year-old children). As previously suggested, one key purpose of the present study was to extend previous work on the development of relational framing broadly by examining the development of a range of different relational frames in young children. Previous RFT studies have used a number of different methodologies to examine and train relational framing in young children. One of the most efficient methods is the relational evaluation procedure (REP). This methodology allows participants to report on or evaluate sets of arbitrarily applicable relations defined by various sets of contextual cues. In one of the most impressive examples of the applied educational utility of this technology, Cassidy et al. (2011) and various follow-up studies (e.g., Cassidy et al., 2016; Hayes & Stewart, 2016) have used the REP to assess and train AADRR by presenting statements involving nonsense words juxtaposed with contextual cues (e.g., "CUG is the SAME as DAX," "DAX is the SAME as YIM"), and asking participants (typically children from age 10 to adolescence) questions that required them to derive relations based on those presented statements (e.g., "Is DAX the SAME as CUG?" "Is CUG the same as YIM?"). Training with this methodology is extremely efficient because multiple exemplars of relational patterns can be easily generated and readily presented, and indeed training with this variation of the REP protocol has been shown to substantially boost children's intellectual performance.

Given that a key goal of the present study was to assess relational framing across various levels, it was decided to employ a "relational statement" format similar in some respects to that used by Cassidy et al. (2011). There was one critical difference, however. In Cassidy et al., participants were required to read sentences specifying and/or querying the relations between nonsense words (e.g., "CUG is the same as DAX"). Considering the range of ages in the present study (i.e., from 3 to 7 years), it seemed probable that some participants, especially the younger ones, were not yet readers or had minimal reading skills. Hence, in order to allow for minimal reading skills, and thus ensure that all participants could be assessed and trained equally effectively on AADRR, the stages in the relational assessment focused on this repertoire utilized colored shapes, single letters, and audio options. It was envisaged that use of this format would allow us to draw on the advantages of the REP (facilitating more efficient and extensive testing of AADRR than MTS) with a younger set of participants.

The training and testing format utilized in the current study was employed in the context of the multistage relational assessment that allowed testing of a range of different types of relations. Also, in accordance with the broader aims of examining the development of relational framing more broadly, it did so at a number of different levels of complexity, including (1) nonarbitrary relations, (2) relating of nonarbitrary relations, (3) arbitrary relations, and (4) relating of arbitrary relations. In addition, participants' relational performance across and within all levels and frames was correlated with their age and intellectual performance, as assessed on a standardized test of intellectual functioning, namely the Stanford-Binet Intelligence Scales (5^{th} edition) for Early Childhood (SB5).

Method

Participants and Settings

Participants included 24 students (14 females, 10 males) attending a private, nonsecular U.S. school in New Jersey. Demographic information for the participants is summarized in Table 1. Participant ages ranged from 36 to 84 months (M = 59.96, SD = 13.76), including five 3- to 4-year-old (M = 39.8 months, SD = 3.9), six 4- to 5-year-old (M = 53.33 months, SD = 2.73), six 5- to 6-year-old (M = 65.67 months, SD = 3.08), and seven 6- to 7-year-old participants (M = 75.14 months, SD = 4.18). None of the participants had any known developmental or intellectual disabilities. Full-scale IQ scores, which were obtained for the purposes of this study, ranged from 94 to 132 (M = 113.25, SD = 10.59).

The assessment measures were administered by the researcher (the first author of the present study) at the participants' school in a separate, quiet classroom within the school building. When conducting the assessments, the researcher and the participant were seated at a child-sized school desk; the participant sat facing the desk and the researcher sat at the side of the desk facing the participant and with a full view of the testing materials in front of the participant.

Materials and Apparatus

The materials used in this study included a standardized measure of intelligence, the Stanford-Binet Intelligence Scales-5^{th} Edition for Early Childhood (SB5; Roid, 2003), and the relational assessment. The relational assessment was presented on a computer; Stages 1 and 2 of the assessment were presented on an Apple iPad Air 2 using Apple Keynote software, and Stages 3 and 4 were presented on an Acer Chromebook R 11 using Microsoft PowerPoint software.

The Stanford-Binet Intelligence Scales-5^{th} Edition for Early Childhood

The SB5 is an assessment of intelligence and cognitive abilities and is considered the standard measure of global intellectual ability for children and adults aged 2–85+ (Roid, 2003). In order to provide a reliable profile of differential abilities, the SB5 yields three composite scores, including Full Scale IQ (FSIQ), Verbal IQ (VIQ), and Nonverbal IQ (NVIQ). Full Scale IQ, VIQ, and NVIQ composite scores have high reliability coefficients ranging from .95 for VIQ to .98 for FSIQ

Table 1 Participant demographic information

		N	%	FSIQ	*SD*
Gender	Male	10	42		
	Female	14	58		
Age (months)	36–47	5	21	118.60	12.76
	48–59	6	25	110.83	9.00
	60–71	6	25	114.67	4.37
	72–84	7	29	110.29	14.01

(Roid, 2003). The validity of the SB5 is supported based on its correlation with a number of alternative intelligence tests including the Wechsler Preschool and Primary Scale of Intelligence-Revised (WPPSI-R), and the Woodcock-Johnson III Tests of Cognitive Abilities-General Intellectual Ability-Standard (WJ III GIA; Garred & Gilmore, 2009). In the present study, in order to calculate correlations between the nonnorm-referenced relational assessment scores and SB5 scores, raw scores were used for SB5 total, Verbal, and Nonverbal scores.

The Relational Assessment

The relational assessment examined different patterns of relational responding across four stages, including Stage 1: nonarbitrary (physical) relations; Stage 2: nonarbitrary analogical relations (relations between physical relations); Stage 3: arbitrary (abstract) relations (relational frames); and Stage 4: arbitrary analogical relations (relations between abstract relations). Within each stage were five substages focused on particular relations, including coordination, comparison, opposition, temporality, and hierarchy. Each substage included 10 trials (see Figure 1 for a schematic presentation of the relational assessment). In what follows, we present the general layout of the stimuli for each of the stages and substages in the assessment, and in the procedure section we will describe the administration of the assessment. In Stages 1 and 2 of the relational assessment, the sample stimulus (in Stage 2, a compound stimulus) was presented at the top of the computer screen, and two comparison stimuli (in Stage 2, compound stimuli) were presented at the bottom left and right of the screen (for an illustrative example, see Appendices A1 and A2). In Stage 3, a relational network of stimuli was presented in the middle of the screen (see Appendix A3), whereas in Stage 4 a relational network of stimuli was presented at the top of the screen, a sample compound stimulus was in the middle of the screen, and the comparison compound stimuli were at the bottom left and right of the screen (see Appendix A4). For all trials, stimuli were delineated with a solid black line including a horizontal black line underneath the sample and a vertical line between the comparison stimuli.

Stage 1: Nonarbitrary Relations The relational stimuli included simple, monochromic pictures or shapes in the coordination, comparison, opposition, and temporal substages, and boxes of varied sizes and colors in the hierarchy substage. The coordination substage included MTS trials of a sample and three comparison stimuli. The comparison substage included trials in which the participant was shown a sample and three comparisons; the comparisons were identical to the sample except for size, and included one bigger, one smaller, and one identical comparison. The opposition substage included trials in which the participant was shown a sample and three comparisons identical to the sample except for variance on one dimension on a gradient scale. For example, if the sample was a black square, the comparisons included a white, gray, and black square. The temporality substage included trials in which the participant was instructed to watch the iPad screen for sequentially appearing shapes. The first stimulus appeared 0.5 s after the onset of a new trial, and the second stimulus appeared 1.0 s after the first stimulus. The hierarchy substage included trials in which the participant was shown two or three different sized and different colored boxes.

Stage 2: Nonarbitrary Analogical Relations In Stage 2, all stimuli presented were compound stimuli. In all substages, a sample compound stimulus was presented at the top of the screen, and two comparison compound stimuli were presented below the sample. The stimuli used in the sample compound were always different from the stimuli used in the comparison compounds. In the coordination substage, the sample was composed of either two identical or two nonidentical pictures, and the comparisons included compound stimuli composed of identical stimuli and nonidentical stimuli. In the comparison substage, each of the sample and comparison compounds included three stimuli that were similar except for size. Two of the three stimuli in each compound were outlined in red, and an arrow indicated which stimulus to compare with another stimulus in the compound (e.g., from smaller to bigger or bigger to smaller). In the opposition substage, each of the sample and comparison compounds included three stimuli that varied on a gradient scale along one dimension (e.g., size, color, quantity). In each compound, two of the stimuli had a solid red line underneath and were either nonarbitrarily opposite or not opposite. In the temporality substage, both the sample and comparison compounds included two stimuli which appeared onscreen either simultaneously or 0.5 s apart. In the hierarchy substage, each trial included sample and comparison

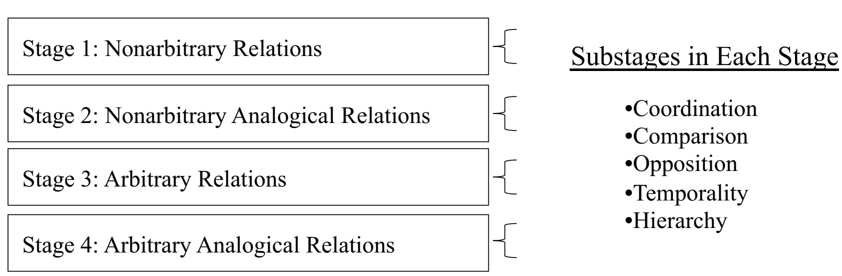


Fig. 1 Schematic representation of the relational assessment stages and substages. *Note.* This figure illustrates the elements and sequence of the relational assessment

stimuli consisting of a square inside another square, and small, blue dots located in either the innermost or outermost square or outside the squares.

Stage 3: Arbitrary Relations An adaptation of the REP format was employed in all substages. The relational stimuli were simple, monochromic shapes (circles, triangles, squares) separated by a single Latin letter indicating the contextual cue (*S* for Same, *D* for Different, *M* for More, *L* for Less, *O* for Opposite, *B* for Before, *A* for After, *C* for Contains, and *I* for Inside), plus corresponding audio icons (which, when touched, produced an audio recording of the contextual cue (e.g., "same") through the laptop or computer speaker).

Stage 4: Arbitrary Analogical Relations The REP format was also used in Stage 4, including the use of monochromic circles for the relational stimuli plus visual and audio signals representing the contextual cues. The sample and comparison compounds were combinations of the relata in the relational network.

Procedure

At the start of the session, the researcher placed the iPad or computer in front of the participant. Participants were provided with detailed instructions before they commenced with each substage in every stage. Responses were scored as either correct or incorrect. A correct response was defined as touching or pointing to the correct comparison or providing a correct vocal response. An incorrect response was defined as touching or pointing to the incorrect comparison, touching both correct and incorrect comparisons, providing an incorrect vocal response, not responding, or engaging in other behavior that could not be categorized as correct. Participants did not receive feedback for any responses during the relational assessment or SB5 testing. Generalized reinforcement was provided for compliance and participation throughout the assessments.

The sequence in which the two assessments were administered was randomized so that participants completed either the relational assessment or the SB5 first. The SB5 was completed in one session, and the relational assessment was completed in one to two sessions. For the relational assessment, each session lasted between 20 and 60 min, including scheduled breaks every 10–15 min, or if the participant requested a break. For all participants, both assessments were conducted within a maximum of 4 weeks of each other.

Prior to conducting the study, ethical approval for recruitment of participants was obtained from the research ethics committee of the host (research) institution. Consent for conducting the study was obtained from the principal of the New Jersey school. Caregiver consent was obtained for each child who participated, and verbal consent was also obtained from each of the participants.

Stage 1: Nonarbitrary Relations

In the coordination substage, participants were to match the sample to a comparison upon hearing the instruction, "Which one of these at the bottom is like this one at the top?" In the comparison substage, participants were asked to identify the correct comparison in relation to the sample upon hearing the instruction, "Which one of these is *bigger* or *smaller* than the one at the top?" In the opposition substage, participants were asked to identify the correct comparison in relation to the sample upon hearing the instruction, "Which one of these is *opposite* to the one at the top?" In the temporality substage, participants were to identify the order in which the stimuli appeared upon hearing instructions such as, "Which one was *before* or *after* the other one/was stimulus 1 *before* or *after* stimulus 2?" In the hierarchy substage, participants were to identify where a box was in relation to another box upon hearing instructions such as, "Which box is inside/which box *contains* the other box?"

Stage 2: Nonarbitrary Analogical Relations

In Stage 2, all stimuli presented were compound stimuli; in all substages, the participants' task was to match the sample (compound stimulus) relation to the correct comparison (compound stimulus) relation. For example, in the coordination substage, the sample was composed of either two identical or two nonidentical pictures, and the comparisons included a compound stimulus composed of two identical stimuli and a compound composed of two nonidentical stimuli. Participants were required to match the sample to one of the comparisons upon hearing the instruction, "Look at these at the top. Which one of these (point to the comparison stimuli) is like the ones at the top?" A pretrial sample was presented before administering each substage in order to familiarize the participant with the testing format (for an illustrative example, see Appendix A2).

Stage 3: Arbitrary Relations

Prior to starting Stage 3, a pretest was administered to familiarize participants with the test format. The participant was shown a computer screen with a relational network, for example: [Red Circle] [S] [Blue Circle]. The assessor instructed the participant to look at the screen and said, "The S means *same*. If you can't remember what the S means, you can tap on the S and the computer will tell you." Once the participant was comfortable with the visual and audio stimuli the assessor provided the instruction, "Let's start. We are going to pretend these shapes like food, and we are going to talk about whether they like or do not like the same food. Look at the screen in front of you." A relational network was presented on the screen, for example: [Red Circle] [S] [Blue Circle]; [Blue Circle] [S] [Yellow Circle], and the assessor read, "Red likes the same food as Blue, and Blue likes the same food as Yellow." The assessor asked questions about the relational network, including questions about directly given relations (e.g., "Does Red like the same food as Blue?"), questions requiring mutual entailment (e.g., "Does Red like different food to Blue?"), and questions requiring combinatorial entailment (e.g., "Does red like the same food as yellow?"). The pretest included 10 yes/no questions.

For all substages in Stage 3, the assessor first read the relational network to the participant and then asked the trial questions. Questions became increasingly difficult and required responses including directly trained, mutually entailed, and combinatorially entailed relations. The coordination substage was like the pretest but introduced novel stimuli.

In the comparison substage, the relational network included colored circles and the contextual cues *more* (M) and *less* (L). Participants were told that they were going to pretend-buy their favorite food or candy, "These circles are like coins and we are going to pretend that you can use them to buy your favorite food/candy."[1] For example, the participant saw: [Blue Circle] [L] [Red Circle] and the assessor said: "Blue buys less than Red, so which coin should you take to the store to buy [insert participant's favorite food]?" The correct choice, in this case, would have been to select the red coin because red buys more.

In the opposition substage, the relational network included the contextual cue (O) for *opposite* and colored circles representing coins. Participants were told that they were going to pretend-buy their favorite food or candy: "These circles are like coins and we are going to pretend that you can use them to buy your favorite food/candy." For example, the participant saw: [Red Circle] [O] [Blue Circle] and the assessor said, "Red buys many/a lot/a few/a little. Red is opposite to Blue, which coin should you take to the store to buy [insert participant's favorite food]?" The correct choice on every trial was the selection of the color worth a lot/many.

In the temporal substage, the relational network included colored squares and the contextual cues *before* (B) and *after* (A). The assessor told the participant that the colored squares were in a race: "These silly little squares are racing, which one reaches the finish line before/after the other one?" For example, the participant saw: [Red Square] [B] [Blue Square] and the assessor said, "Red was before Blue. Was Blue before/after Red?"

In the hierarchy substage, the relational network included colored circles and the contextual cues *inside* (I) and *contains* (C). The assessor told the participant that the colored circles either contained each other or were inside each other. For example, the participant saw: [Red Circle] [I] [Blue Circle] and the assessor said, "Red is inside Blue." The assessor would then present one of the following questions depending on the particular trial-type: Is Blue inside Red? Does Blue contain Red? Does Red contain Blue? Which one is inside? Which one contains the other one?

Stage 4: Arbitrary Analogical Relations

The same relational network composed of circles and single letters was used for all trials in each substage. The sample and comparison compounds were combinations of the relata in the relational network. For example, in the coordination substage, the relational network would be read as, "Blue is the same as yellow, yellow is the same as red, and red is different to green." The sample compound may be [Blue : Red] and the comparison compounds may be [Yellow : Green] and [Red : Yellow] (see Appendix A4 for an illustrative example for each of coordination, comparison, opposition, temporality, and hierarchy). On each trial, the researcher read the relational network to the participant and then delivered the instruction, "Look at this one at the top (pointing to the sample compound). Which one of these (pointing to each of the comparison compounds in turn) is like this one at the top?" For example, given the relational network mentioned above, [Blue S Yellow], [Yellow S Red], and [Red D Green], participants could derive the relation of sameness for a sample stimulus [Blue : Red], a relation of sameness for a comparison stimulus [Red : Yellow], and a relation of difference for a comparison stimulus [Yellow : Green]. This example was for the coordination substage; for each of the other substages, the participants had to match the correct comparison relation to the appropriate sample relation, but the nature of the relations that had to be matched depended on which substage it was. There were 10 trials in each substage, including trials for directly trained, mutually entailed, and combinatorially entailed relations in that order (i.e., increasing difficulty).

Interobserver Agreement

Interobserver agreement was calculated for 20% of all completed relational assessment substages by a trained research assistant. The research assistant was trained in data collection until they reached 100% accuracy prior to collecting IOA data. Trial-by-trial IOA was calculated and ranged from 97% to 100% (M = 98.66%). Agreement across all measures was counted if both observers provided the same score for each item assessed, and disagreement was counted if one observer provided a different score compared to the other observer. Interobserver agreement was calculated for 24% of the verbal (Vocabulary) and nonverbal (Object Series/Matrices) routing substages of the SB5. Trial-by-trial IOA was calculated and

[1] Adaptation of the procedure used in Barnes-Holmes et al. (2004b).

ranged from 94% to 100% ($M = 97.83\%$). Treatment integrity was evaluated on 20% of all completed relational assessment substages, and 23.5% of all SB5 routing substages by a trained research assistant. The observer scored a + on each trial that the trainer was 1) observed to gain attention prior to the trial, 2) accurately presented the question, and 3) consequated appropriately. Mean procedural integrity ranged from 97% to 100% ($M = 99\%$) for the relational assessment substages, and from 88% to 100% (*M = 98.1%)* for the SB5 routing substages.

Results

In the results section we will examine the relational assessment protocol scores, correlations between relational assessment scores and age, correlations between relational assessment scores and raw IQ, correlations within the protocol, and finally, the acquisition of relations across age groups. Results for Stages 2 (nonarbitrary analogy) and 4 (arbitrary analogy) will receive particular attention in each subsection. In addition, because percentage of correct responding is provided as a performance outcome across different trial types, in Appendix 2 we provide a table showing chance level of responding (based on possible number of outcomes) for different trial types. In the section below, in which we examine performance across relational frames, we compare performance of the different cohorts on different stages and for different relations with chance level performance.

Relational Assessment Protocol Scores

Table 2 shows scores on both the relational assessment (overall and by stage) and the SB5 (full scale and subscale) both as a function of the entire cohort as well as by age group. Total relational assessment scores across all participants ranged from 26.5% to 78% correct ($M = 56\%$, $SD = 28.58$). Stage 1 (nonarbitrary relations) scores across all substages and all participants ranged from 58% to 94% ($M = 80\%$, $SD = 5.19$); Stage 2 (nonarbitrary analogy) scores ranged from 40% to 84% ($M = 61\%$, $SD = 5.46$); Stage 3 (arbitrary relations) scores ranged from 0% to 88% ($M = 54\%$, $SD = 10.70$); and Stage 4 (arbitrary analogy) scores ranged from 0% to 66% ($M = 29\%$, $SD = 12.53$).

Relational Assessment Protocol Scores and Age

With regard to breakdown by age group, we can see relatively predictable patterns wherein relational assessment scores increased as a function of age and mean relational assessment scores increased by approximately 10% for each additional year in age. In particular, total relational assessment scores ranged from 26.5% to 52% correct ($M = 38\%$, $SD = 19.84$) for the 3- to 4-year-old cohort, from 40% to 58% correct ($M = 48\%$, $SD = 12.15$) for the 4- to 5-year-old cohort, from 50.5% to 71.5% correct ($M = 60.4\%$, $SD = 16.62$) for the 5- to 6-year-old cohort, and from 68% to 78% correct ($M = 71\%$, $SD = 7.3$) for the 6- to 7-year-old cohort.

Mean scores within each stage showed improvements across age cohorts. The biggest improvement in relational assessment scores based on age occurred in the later stages, that is, Stages 3 (arbitrary relations) and 4 (arbitrary analogy). For example, comparing the youngest with the oldest cohorts, we see that in Stage 3, the 3- to 4-year-old cohort scored on average 20% correct across all substages whereas the 6- to 7-year-old cohort scored 75% correct, a difference of 55%. In Stage 4, the 3- to 4-year-old cohort scored on average 10% correct whereas the 6- to 7-year-old cohort scored 54% correct, a difference of 44%. In contrast, in Stages 1 and 2, there was less of an age-based gap in performance. In Stage 1, the 3- to 4-year-old cohort scored an average of 72% correct whereas the 6- to 7-year-old cohort scored 86% correct, a difference of 14%, whereas in Stage 2, the 3- to 4-year-old cohort scored an average of 50% correct whereas the 6- to 7-year-old cohort scored 68% correct, a difference of 18%.

Apart from considering the general improvement based on age, we can also examine each stage to check for particular discontinuities between specific age cohorts. For example, in Stage 2 (nonarbitrary analogy), there is a large gap in performance between the 4- to 5-year-old cohort and 5- to 6-year-old cohort (M = 54% to 68%, respectively). These data show that the transition from the 4- to 5-year-old to the 5- to 6-year-old age range results in particular improvement in nonarbitrary analogy. In Stage 3 (arbitrary relations), the biggest improvement in performance occurred between the 3- to 4-year-old and 4- to 5-year-old participants (M = 20% to 56%, respectively). These data suggest substantial development in arbitrary relational framing between these ages. In Stage 4 (arbitrary analogy), the biggest improvement in performance occurred between the 4- to 5-year-old and 5- to 6-year-old cohorts *(M* = 8% and 34%, respectively). This finding for arbitrary analogy is similar to that for nonarbitrary analogy in Stage 2. The fact that there appears to be substantive improvement in analogical ability between the 4- to 5-year-old and 5- to 6-year-old participants at both the nonarbitrary and arbitrary levels supports earlier research findings (e.g., Carpentier et al., 2002) that analogical ability begins to emerge around 5 years of age. One final point concerns the difference in variability in performance in arbitrary analogy between different cohorts. In particular, the standard deviation decreased from 11.37 for the participants aged 5- to 6-years-old to 4.3 for participants aged 6- to 7-years-old. This suggests less variable, more stable responding in the older cohort following the initial emergence of analogy.

Table 2 Participant scores on the relational assessment and SB5

ID	M/ F	Age (m)	SB5- FSIQ	SB5- NV	SB5- V	SB5- T	RA- T	RA1 NAR	RA2 NAA	RA3 AR	RA4 AA
3- to 4-year-old cohort											
S4	F	36	127	46	57	103	62	29	24	9	0
S26	F	37	132	54	59	113	104	35	26	17	26
S3	F	40	100	37	40	77	53	33	20	0	0
S5	F	40	122	55	48	103	77	39	28	10	0
S17	F	46	112	55	50	105	84	43	27	14	0
Mean		39.80	118.60	49.40	50.80	100.20	76.00	35.80	25.00	10.00	5.20
SD		3.90	12.76	7.89	7.60	13.61	19.84	5.40	3.16	6.44	11.63
4- to 5-year-old cohort											
S11	M	52	110	59	60	119	80	32	26	22	0
S22	M	55	102	50	62	112	102	43	26	29	4
S23	M	49	126	62	68	130	116	42	34	31	9
S27	F	53	103	51	57	108	93	37	24	28	4
S34	F	54	108	54	66	120	95	38	27	30	0
S60	F	57	116	74	64	138	90	31	24	28	7
Mean		53.33	110.83	58.33	62.83	121.17	96.00	37.17	26.83	28.00	4.00
SD		2.73	9.00	8.96	4.02	111.18	12.15	4.96	3.71	3.16	3.63
5- to 6-year-old cohort											
S39	M	68	118	82	80	162	134	41	42	34	17
S40	F	61	110	64	76	140	105	38	32	31	4
S46	F	63	121	78	76	154	143	46	37	27	33
S47	M	67	115	79	69	148	101	41	31	25	4
S49	M	66	110	73	67	140	115	43	29	22	21
S59	M	69	114	81	77	158	127	46	32	26	23
Mean		65.67	114.67	76.17	74.17	150.33	120.83	42.50	33.83	27.50	17.00
SD		3.08	4.37	6.74	5.04	9.24	16.62	3.15	4.79	4.32	11.37
6- to 7-year-old cohort											
S52	F	75	94	76	69	145	136	40	32	27	27
S53	M	76	123	87	101	188	156	47	40	41	28
S54	M	72	128	86	99	185	138	37	33	35	33
S55	F	74	100	74	70	144	138	42	32	38	26
S58	F	72	95	74	63	137	136	47	35	35	19
S56	M	73	120	80	92	172	144	43	30	44	27
S61	F	84	112	93	92	185	146	46	37	33	30
Mean		75.14	110.29	81.43	83.71	165.14	142.00	43.14	34.14	37.57	27.14
SD		4.18	14.01	7.39	15.83	22.37	7.30	3.80	3.44	3.82	4.30
All participants											
Mean		59.96	113.25	67.67	69.25	136.92	111.46	39.96	30.33	26.92	14.25
SD		13.76	10.59	14.97	15.46	29.25	28.58	5.19	5.46	10.70	12.53

SB5-FSIQ Stanford-Binet Intelligence Scales-5th Edition for Early Childhood Full Scale IQ; *NV* nonverbal; *V* verbal, *T* total, *RA* relational assessment, *NAR* nonarbitrary relations, *NAA* nonarbitrary analogical relations, *AR* arbitrary relations, *AA* arbitrary analogical relations

Correlating Assessment Protocol Performance and Age

Figure 2 shows graphs of the relationships between scores on each of the four relational assessment stages and age. These data suggest strong correlations between each of the stages and age. Table 3 shows a matrix of Spearman's rank correlations between variables including age, SB5 total and substage raw scores, and relational assessment total and substage scores. There is a strong correlation between relational assessment total score and age (r = .859, $p < 0.01$). Further analyses also show strong, significant correlations between total scores for each stage and age (Stage 1: r = .604, p < 0.01; Stage 2: r = .709, p < 0.01; Stage 3: r = .818, p < 0.01; and Stage 4: r = .746, p < 0.01). All correlations across stages and age were significant at the .01 level. These correlations suggest, consistent with RFT, that the capacities to engage in relational responding and analogical responding (both overall as well as across different frames) are established and strengthened via ongoing exposure to the typical socioverbal environment.

Within-stage analyses show a significant correlation between age and comparison relations in Stage 1, and age and coordination, and hierarchy in Stage 2. These analyses show significant correlations between age and all substages in Stages 3 and 4.

Relational Assessment Protocol Performance and Raw IQ

Figure 3 shows graphs of the relationships between scores on each of the four relational assessment substages and raw IQ scores. Spearman's rank analysis reveals a strong correlation between relational assessment total scores and raw IQ scores (r = .874, $p < 0.01$). Further analyses show strong correlations between each stage of the relational assessment scores and raw IQ scores (Stage 1: r = .512, $p < 0.05$; Stage 2: r = .782, p < 0.01; Stage 3: r = .717, $p < 0.01$; and Stage 4: r = .815, $p < 0.01$). Furthermore, each stage of the relational assessment also correlated with both SB5-NV and SB5-V subscale raw scores (SB5-NV and Stage 1: r = .534, $p < 0.01$; Stage 2: r = .806, $p < 0.01$; Stage 3: r = .637, $p < 0.01$; and Stage 4: r = .760, $p < 0.01$; and SB5-V and Stage 1: r = .484, $p < 0.05$; Stage 2: r = .788, $p < 0.01$; Stage 3: r = .753, $p < 0.01$; and Stage 4: r = .774, $p < 0.01$). Overall, these correlations suggest, consistent with RFT, a strong link between relational responding (both overall as well as across different frames) and intellectual potential, as reflected in the raw IQ score.

Within-stage analyses show a significant correlation between Stage 1 comparison and hierarchy scores, and SB5 raw score. These findings are consistent with previous research showing the importance of comparative relational framing for intellectual potential (Cassidy et al., 2011, 2016). There are also significant correlations between Stage 2 coordination, comparison, and hierarchy scores, and SB5 raw score. Finally, in Stages 3 and 4, there are significant correlations between all substage scores and SB5 raw scores. This pattern is consistent with the idea that arbitrary relational framing is particularly important for intellectual potential. Upon further examination across all four stages, comparison and hierarchy are the only substages that show a consistent correlation with raw IQ as revealed by both SB5 total score as well as SB5 subscale scores. This pattern suggests that comparison and hierarchy may be particularly important foundational intellectual skills.

Correlating Intraprotocol Relations

Spearman's rank analysis in Table 3 shows strong, significant relationships within and across many of the stages and substages. Stage 1 comparison and hierarchy scores show significant correlations across substage total scores for all stages; these data may suggest that nonarbitrary comparison and hierarchy are important prerequisite relations for more complex language. Stage 2 coordination scores show significant correlations across substage total scores for all stages; these data indicate that performance on nonarbitrary coordinate analogy may be a good predictor of relational responding more generally. Stage 3 coordination and comparison scores show significant correlations across substage total scores for all stages whereas Stage 4 coordination, comparison, and opposition scores show significant correlations across substage total scores for all stages. Again, this seems to indicate that arbitrary relational framing may be particularly important for intellectual potential in general.

Only the comparison substage in Stage 1 shows significant correlations with age, IQ, and total relational assessment score. The coordination and hierarchy substages in Stage 2 show significant correlations with age, IQ, and total relational assessment score. All Stage 3 and Stage 4 substages show strong, significant correlations with age, raw IQ score, and total relational assessment score. These findings further support the key importance of arbitrary relational responding (i.e., relational framing) as a core intellectual ability that is acquired across childhood.

In both Stages 1 and 3, the comparison substage is strongly correlated with all of the substages in Stage 4. These data suggest nonarbitrary and arbitrary comparative relational repertoires are important intellectual skills, and are strong predictors of analogical ability. Stage 1 comparison scores also show a significant correlation with Stage 2 coordination scores, which in turn, as previously noted, shows significant correlations with age, IQ, and total relational assessment score. Thus, it may seem that having a strong nonarbitrary comparative repertoire facilitates nonarbitrary coordinate analogy, which supports nonarbitrary and arbitrary relational responding.

Examining Performance Across Relational Frames

One final analysis looked at the acquisition of relational framing in each stage. Figure 4 shows the mean percent correct for each relational frame across stages, and for each age cohort.

In general, in Stage 1 (nonarbitrary relations), there is an improvement with age across cohorts. Performance across cohorts shows little variation in coordination (mean scores range from 98% to 100% correct).

Table 3 Matrix of Spearman's Rho correlations for all measures administered

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29
1. Age	–																												
2. SB5-T	.879**	–																											
3. SB5-NV	.856**	.958**	–																										
4. SB5-V	.832**	.973**	.894**	–																									
5. RA-T	.859**	.874**	.810**	.871**	–																								
6. S1-T	.604**	.512*	.534**	.484*	.688**	–																							
7. S2-T	.709**	.782**	.806**	.788**	.831**	.684**	–																						
8. S3-T	.818**	.717**	.637**	.753**	.817**	.430*	.649**	–																					
9. S4-T	.746**	.815**	.760**	.774**	.915**	.504*	.672**	.639**	–																				
Stage 1 Substages: 1 = Coordination; 2 = Comparison, 3 = Opposition; 4 = Temporality; 5 = Hierarchy																													
10. SS-1	0.316	0.349	0.327	0.371	.436*	0.394	0.394	0.381	0.396	–																			
11. SS-2	.548**	.595**	.578**	.534**	.625**	.516**	.477*	.439*	.631**	.416*	–																		
12. SS-3	0.332	0.225	0.242	0.206	.429*	.821**	.459*	0.28	0.224	0.265	.474*	–																	
13. SS-4	0.393	0.313	0.355	0.277	0.357	.528**	0.252	0.122	0.321	0.121	−0.084	0.178	–																
14. SS-5	0.384	.431*	.430*	.487*	.577**	.564**	.767**	.457*	.470*	.483*	0.165	0.294	0.211	–															
Stage 2 Substages: 1 = Coordination; 2 = Comparison, 3 = Opposition; 4 = Temporality; 5 = Hierarchy																													
15. SS-1	.605**	.702**	.693**	.680**	.648**	.422*	.714**	.429*	.657**	0.333	.590**	0.151	0.212	.475*	–														
16. SS-2	0.326	.499*	.469*	.541**	.477*	.558**	.592**	0.318	0.31	0.269	0.299	.434*	0.156	.527**	0.235	–													
17. SS-3	0.211	0.289	0.295	0.318	0.288	.446*	.556**	0.154	0.085	0.238	−0.125	0.322	0.259	.563**	0.17	0.343	–												
18. SS-4	0.329	0.154	0.171	0.134	0.262	0.389	0.355	0.226	0.114	0.123	0.199	.418*	0.069	0.262	0.089	−0.041	0.335	–											
19. SS-5	.568**	.582**	.637**	.557**	.683**	0.374	.690**	.641**	.657**	0.154	0.376	0.173	0.185	.456*	.612**	0.221	0.049	−0.047	–										
Stage 3 Substages: 1 = Coordination; 2 = Comparison, 3 = Opposition; 4 = Temporality; 5 = Hierarchy																													
20. SS-1	.658**	.605**	.555**	.590**	.682**	.643**	.621**	.687**	.521**	.419*	.526**	.625**	0.121	.432*	.427*	0.346	0.359	0.237	.551**	–									
21. SS-2	.730**	.715**	.631**	.693**	.837**	.421*	.525**	.754**	.826**	.419*	.687**	0.175	0.135	0.286	.474*	0.217	0.008	0.196	.513*	.455*	–								
22. SS-3	.568**	.455*	.458*	.470*	.554**	0.298	.579**	.718**	.428*	0.274	0.17	0.116	−0.004	.571**	0.312	0.242	0.338	0.246	.493*	0.362	.533**	–							
23. SS-4	.676**	.684**	.606**	.704**	.632**	0.185	.546**	.781**	.480*	0.033	0.209	0.031	0.065	0.28	0.356	0.392	0.144	0.101	.530**	.416*	.506*	.569**	–						
24. SS-5	.640**	.560**	.436*	.620**	.567**	0.229	0.31	.812**	.442*	0.378	0.345	0.194	0.147	0.19	0.318	0.098	−0.083	0.122	0.353	.586**	.512*	0.336	584**	–					
Stage 4 Substages: 1 = Coordination; 2 = Comparison, 3 = Opposition; 4 = Temporality; 5 = Hierarchy																													
25. SS-1	.703**	.749**	.667**	.729**	.829**	.551**	.548**	.557**	.865**	0.399	.496*	0.262	.453*	0.347	.576**	0.341	0.114	−0.01	.530**	.444*	.741**	0.375	.450*	.432*	–				
26. SS-2	.702**	.685**	.678**	.618**	.850**	.579**	.661*	.595**	.885**	0.339	.703**	.409*	0.23	0.348	.612**	0.273	0.055	0.124	.731**	.629**	.734**	.405*	.428*	0.315	.751**	–			
27. SS-3	.657**	.776**	.759**	.737**	.864**	.445*	.737**	.613**	.911**	0.317	.546**	0.221	0.222	.458*	.610**	0.403	0.146	0.025	.759**	.489*	.774**	.468*	.480*	0.305	.782**	.837**	–		
28. SS-4	.648**	.658**	.616**	.598**	.769**	0.403	.491*	.476*	.893**	0.262	.599**	0.158	0.296	0.289	.501*	0.127	−0.073	0.22	.505*	0.392	.738**	0.184	0.306	0.356	.708**	.792**	.768**	–	
29. SS-5	.583**	.620**	.573**	.560**	.747**	0.389	.486**	0.403	.894**	0.262	.599**	0.14	0.278	0.308	.529**	0.132	−0.059	0.225	.479*	0.322	.733**	0.172	0.258	0.263	.726**	.779**	.782**	982**	–

Note. ** Correlation is significant at the 0.01 level (2-tailed); * Correlation is significant at the 0.05 level (2-tailed); *SB5* Stanford-Binet Intelligence Scales - 5th Edition for Early Childhood Full Scale IQ; *NV* non-verbal; *V* verbal; *T* total; *RA* relational assessment; *S* stage; *SS* substages

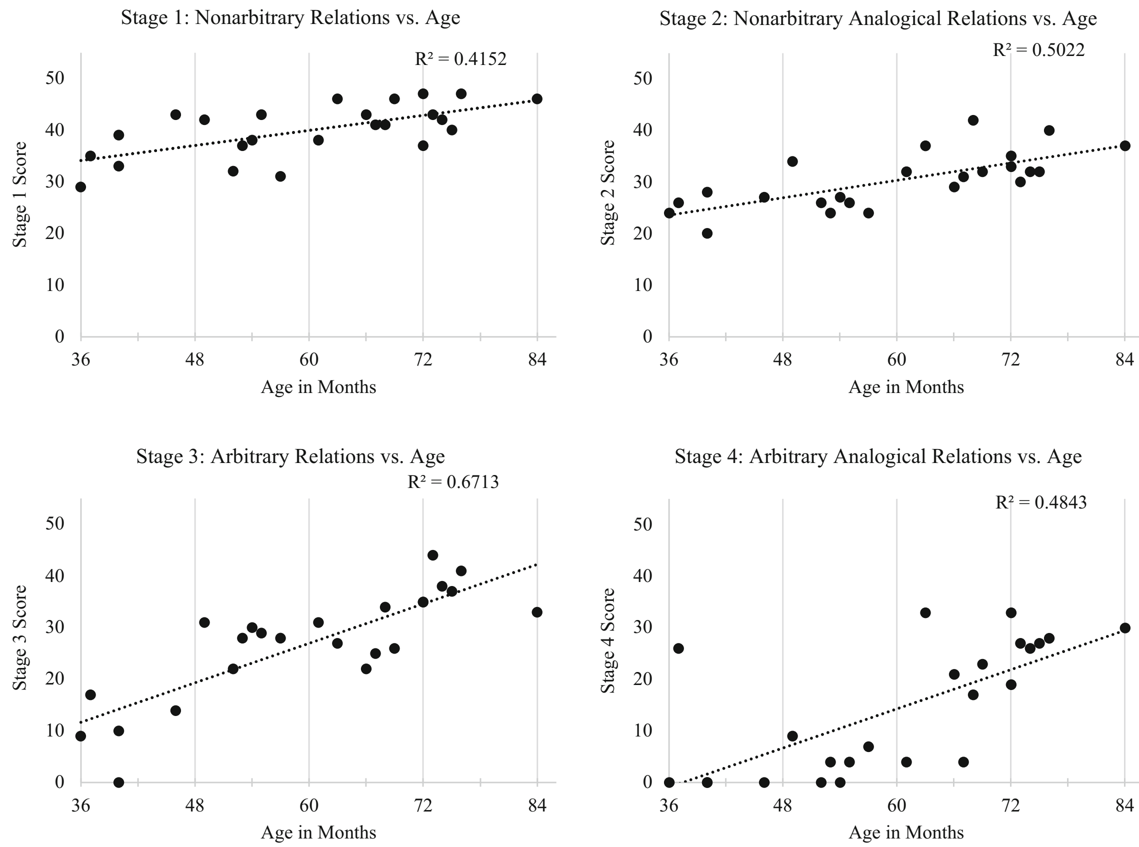


Fig. 2 Relational assessment vs. age across stages. *Note.* Regression slopes are shown for participant scores at each stage and age

Comparative and hierarchical relational performance start to emerge and show steady improvement as age increases, especially between the 4- to 5-year-old and 5- to 6-year-old cohorts. Participants across all cohorts perform lower in temporality and opposition, and performance in both these frames gradually improves across cohorts. Comparing performance across Stage 1 relations with chance level responding (see Figure 4) we can see that the two older cohorts are responding above chance levels on all relations whereas the two younger cohorts are performing above chance levels in all relations except temporality.

In Stages 2 (nonarbitrary analogy) and 3 (arbitrary relations), performance across all relational frames generally improves with age. The most obvious improvement across all frames with age occurs in Stage 3. In particular, there is a clear increase in performance between the 3- to 4-year-old and 4- to 5-year-old cohorts. Comparing performance across Stage 2 relations with chance level responding we can see that the only relation on which all cohorts are responding above chance is sameness. In the case of other relations, the two younger cohorts are at or below chance in the case of all relations,[2] whereas the two older cohorts both show above chance level responding for all other relations except temporality (5- to 6-year-olds) and opposition (6- to 7-year-olds). By showing how, in a slightly unusual context (i.e., nonarbitrary analogy), even older children can still show weak performance on these relational repertoires, the latter pattern perhaps suggests their potential difficulty. Comparing performance across Stage 3 relations with chance level responding, we note first that the youngest group fails to meet a criterion of chance level of responding on any relations whereas the oldest group does so for all relations. The two other groups are somewhere in between with both showing strong performance relative to chance for sameness, comparison, and opposition while showing near chance level responding for both temporality and hierarchy.

As regards analogy in Stage 4 (arbitrary analogy), the 3- to 4-year-old and 4- to 5-year-old cohort scores are low and well below chance level performance. Performance shows substantial improvement in the 5- to 6-year-old cohort and this improvement continues in the 6- to 7-year-old cohort. Furthermore, in the latter group analogical performance is improved not just in coordination but also in various other frames. In particular, the 5- to 6-year-old cohort shows obvious improvement in performance in coordinate analogy. However, participants are still not doing that well (even the coordination performance is only just above chance level), indicating that analogy emerges at this age but it is still a fragile repertoire. The data show gradual improvement by the 6- to 7-year-old cohort, and frames beyond coordination are emerging and improving. It might be noted from the graph that even the oldest cohort are still not responding much above chance level for any of the relations. However, a more detailed

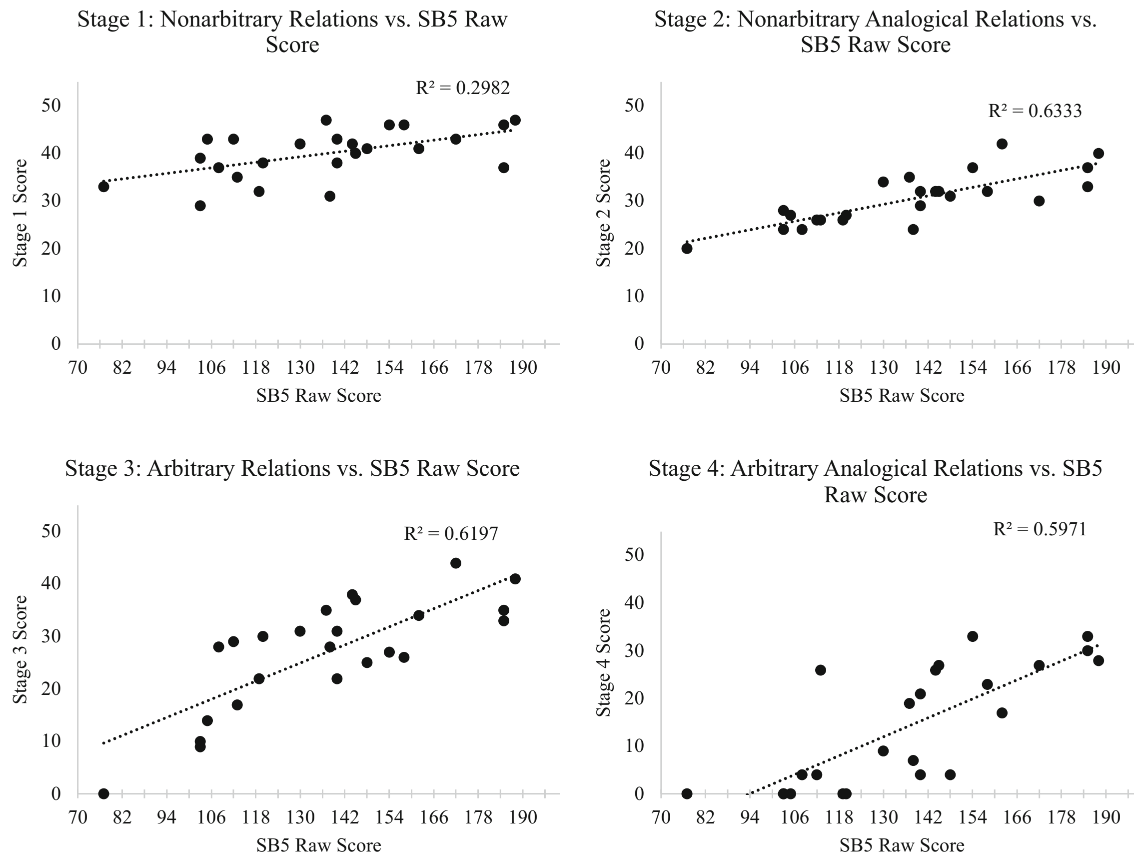


Fig. 3 Relational assessment scores vs. raw IQ score across stages. *Note.* Regression slopes are shown for participant scores at each stage and SB5 raw scores

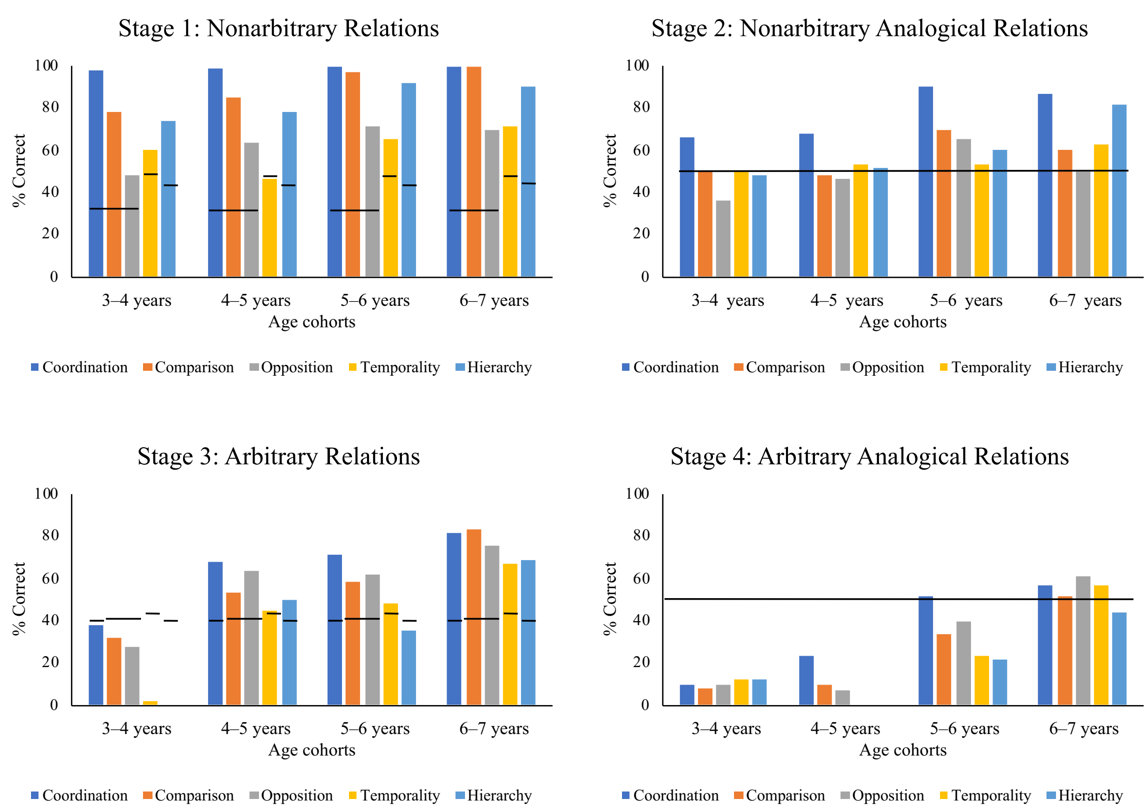


Fig. 4 Mean normative acquisition across stages and relations. *Note.* Mean relational assessment scores across substages and age cohorts for each stage in the relational assessment. The black bars represent chance level responding for each frame across each level

analysis of our data in this respect showed that children in the group were not responding randomly on trials as they tended to answer simpler trials correctly while failing more difficult ones. This pattern was also observed more generally across all stages and all relations, suggesting that the comparison with chance level responding should be seen as a guide rather than indicative of whether children were actually responding at random or not.

It was not a focus of the present study to analyze response differences by gender. However, we performed a cursory and tentative analysis (given the small numbers involved) by gender and found no noticeable differences between females and males on relational assessment scores either within stages or for the assessment as a whole.

Summary

The normative data suggest, consistent with RFT, that the capacity to engage in relational responding and analogical responding (both overall as well as across different frames) are established and strengthened via ongoing exposure to the typical socioverbal environment. Relational performance is correlated with measured IQ which supports the RFT concept that relational framing is critical to language and cognition. Furthermore, the arbitrary stages are more highly correlated with IQ than the nonarbitrary stages. Comparison framing seems to be a particularly important relation as it shows significant correlations with age, IQ, and total relational assessment score. Considering the five relational frames tested in the assessment, several patterns are evident. Nonarbitrary and arbitrary coordination emerges first, and temporality emerges last. Also, there is a difference between nonarbitrary and arbitrary relational responding for frames of opposition and hierarchy; opposition scores are lower in the nonarbitrary stages and higher in the arbitrary stages, and the hierarchy scores are higher in the nonarbitrary stages but lowest in the arbitrary stages.

Discussion

There is now considerable evidence that arbitrary relational responding is the principal behavior characterizing human language and cognition (Dymond & Roche, 2013; Fryling et al., 2020; Zettle et al., 2016). However, despite the evidence, research on the normative development of relational responding, including the sequence of relational frame acquisition, is as yet limited. This study is one of the first attempts to provide a cross-sectional assessment of the acquisition of a selection of frames in young children. Relational responding of coordination, comparison, opposition, temporality, and hierarchy was assessed in 3- to 7-year-old children across four levels of complexity, and performance was correlated with age and intellectual skill as assessed on a standardized IQ test. In addition, a key focus of the relational assessment was analogical responding, a particularly important form of relational framing with respect to intellectual potential (Sternberg, 1977; Sternberg et al., 2001).

The correlations observed in this study between relational performance and IQ score support and extend the growing body of research showing that relational framing supports linguistic and cognitive performance (e.g., Cassidy et al., 2011, 2016; Dixon et al., 2018; Gore et al., 2010; Hayes & Stewart, 2016; Moran et al., 2014; Mulhern et al., 2017; O'Hora et al., 2005, 2008; O'Toole & Barnes-Holmes, 2009). The results in this study also showed significant correlations between relational assessment scores and age, adding to the extant research showing that relational responding strengthens as a function of age in typically developing children (e.g., McHugh et al., 2004; Mulhern et al., 2017). The strong correlations observed in this study indicate that the relational assessment employed here could be a reliable, relatively comprehensive tool for assessing relational skills in young children. Lastly, an important function of this study was to focus in particular on analogical reasoning, and the significant correlations found between analogy and IQ performance provide further evidence that analogy is a particularly relevant pattern of relational framing with regard to intellectual potential (e.g., Sternberg, 1977).

Relational Responding and IQ

Considering these data, first, we see strong correlations between relational ability and IQ (see also Cassidy et al., 2011; Dixon et al., 2018; Gore et al., 2010; Moran et al., 2014; Mulhern et al., 2017; O'Hora et al., 2005, 2008; O'Toole & Barnes-Holmes, 2009). The present study represents an advance in one respect over previous studies that have correlated relational framing and IQ in terms of its scope; we examined derived relational responding across multiple frames and included typically developing children across a number of different age cohorts.

The significant correlations observed in this study between relational performance and IQ support and extend the growing body of research finding that framing relationally is a requisite for linguistic and cognitive performance. In the present study, the correlations between IQ and responding in the arbitrary relational stages were more significant than those between IQ and responding in the nonarbitrary stages. Furthermore, within-stage analyses also showed significant correlations between *all* substage scores and IQ in arbitrary Stages 3 and 4. These correlations suggest, consistent with previous RFT research (e.g., Cassidy et al., 2011, 2016), a strong link between relational framing and intellectual potential. Additional within-stage analyses showed significant correlations between IQ and both nonarbitrary comparison and hierarchy scores in Stage 1. These findings are consistent with previous research noting the importance of comparative relational responding for intellectual potential (Cassidy et al., 2011, 2016). There were also significant correlations between Stage 2 coordination, comparison, and hierarchy scores, and IQ; findings consistent with the literature on analogy and intelligence (Christie & Gentner, 2014; Gentner & Christie, 2010; Goswami & Brown, 1989; Sternberg, 1977). Upon further examination, comparison and hierarchy were the only substages across all four stages that showed a consistent correlation with raw IQ as revealed by both SB5 total score as well as SB5 subscale scores. This pattern suggests that comparison and hierarchy may be particularly important foundational intellectual skills and should be considered for further investigation.

Relational Responding and Age

The relational assessment developed in the present study examined five relational frames, including coordination, comparison, opposition, temporality, and hierarchy at both nonarbitrary and arbitrary levels. Performance on each frame was examined independently within and across stages in an effort to elucidate frame development across age. Results showed relatively predictable patterns wherein total relational assessment scores increased as a function of age, and average scores for each stage showed clear, stable improvements across age cohorts. A strong correlation was found between total assessment score and age, and further analyses showed significant correlations between total scores for each stage and age. These correlations suggest, consistent with RFT, that the capacity to engage in relational responding (both overall as well as across different frames) is established and strengthened via ongoing exposure to the typical socioverbal environment.

The biggest age-based improvement in relational assessment scores between the youngest and oldest cohorts occurred in Stage 3 (arbitrary relations) and Stage 4 (arbitrary analogy). In these two stages, the gap in terms of overall correct responding between the 3- to 4-year-old and the 6- to 7-year-old cohorts was substantial, whereas the difference in scores was not as profound in Stages 1 and 2. These data suggest that there is already substantial development in nonarbitrary relations (as in Stage 1) before the age of 3 years, resulting in relatively less room for improvement after age 3. Apart from finding a general improvement based on age, we also found that there were particular discontinuities between specific age cohorts within stages. For example, in Stage 2 (nonarbitrary analogy), we noted a large gap in performance between the 4- to 5-year-old and 5- to 6-year-old children. These data may indicate that the transition from the 4- to 5-year-old to the 5- to 6-year-old age range results in particular improvement in nonarbitrary analogy. In Stage 3 (arbitrary relations), the biggest age-based improvement in relational assessment scores occurred between the 3- to 4-year-old and 4- to 5-year-old cohorts, and there was almost no change between the 4- to 5-year-old and 5-to 6-year-old cohort. These data might suggest considerable development in arbitrary relational framing between the 3- to 4-year-old and 4- to 5-year-old cohorts. This is an important finding because the development of relational framing as a generalized operant across frames has not previously received much attention. One previous study (Mulhern et al., 2017) investigated the acquisition of hierarchical relations. Mulhern et al. found that participants in the 3- to 4-year-old cohort did not show arbitrarily applicable responding in accordance with hierarchy whereas participants in the 5- to 6-year-old cohort showed arbitrary hierarchical framing relatively robustly. These patterns of development across frames should be considered when designing an RFT-based curriculum focused on training relational framing, a point we discuss further below.

With respect to analogy (relating of relations) in the present study, data from Stage 4 (arbitrary analogy), similar to that from Stage 2 (nonarbitrary analogy), showed that the biggest age-based improvement in relational assessment scores occurred between the 4- to 5-year-old and 5- to 6-year-old cohorts, and there was almost no difference between the 3- to 4-year-old and 4- to 5-year-old cohorts. The fact that there appears to be substantive improvement in analogical ability between the 4- to 5-year-old and 5- to 6-year-old participants at both the nonarbitrary and arbitrary levels supports earlier research (e.g., Carpentier et al., 2002, 2003), which also found an emergence of analogical reasoning around 5 years of age.

Within stage analyses also showed some noteworthy correlations with respect to particular relational frames and age. For example, in Stage 1, the only significant correlation with age was the comparative relation; in Stage 2, coordination and hierarchy correlated with age; and, in Stages 3 and 4, all substages correlated with age. These analyses suggest some interesting findings to which we will return later.

As previously suggested, despite substantial evidence showing that relational framing is a core skill underlying cognitive ability, research on the normative development of relational responding and sequence of frame acquisition is as yet limited. In this study, we investigated the normative development of multiple relational frames in participants aged 3- to 7-years-old. This is one of the first attempts to provide a cross-sectional, comprehensive look at the acquisition of frames in young children.

The relational assessment traced the acquisition of nonarbitrary to arbitrary relational responding and nonarbitrary to arbitrary analogical responding across frames of coordination, comparison, opposition, temporality, and hierarchy. The normative data suggest, in line with RFT, that the ability to engage in relational responding (in general and across different frames) is acquired gradually throughout childhood (at least with respect to typically developing children). The cross-sectional data suggest that typically developing children acquire the coordination relation first. This further supports previous suggestions that the coordination relation is the most common and ubiquitous pattern of relational responding, and that its acquisition may facilitate the development of other relational patterns in a child's developing verbal repertoire (Barnes-Holmes, Barnes-Holmes, & Smeets, 2004; Hughes & Barnes-Holmes, 2016).

The current data also support the importance of comparative responding in accordance with age, IQ, and total relational assessment score. Nonarbitrary comparison was the only nonarbitrary relation that showed significant correlations with age, IQ score, total relational assessment score, and total scores for each stage. These findings suggest that nonarbitrary comparative relations may be an especially relevant relational pattern for intellectual potential, the emergence of arbitrary relational framing, and analogical responding. Furthermore, these data also suggest that the nonarbitrary comparative relation may play an important role in additional language development. Future researchers could further investigate training nonarbitrary comparative relations if they are weak or missing in a child's verbal repertoire in order to facilitate additional language development. The present data also suggest that arbitrary comparison is particularly important; for example, there were significant correlations between age, IQ, and total assessment score and comparison at every stage with the exception of Stage 2 comparison and age. In addition, the comparison data show that arbitrary comparative relations had significant correlations with arbitrary coordination, opposition, temporality, and hierarchy; with total scores for each stage in the assessment; and with all frames in Stage 4, arbitrary analogical responding. These findings provide further evidence that the comparative relation may be especially important for more complex language performance. This also echoes the finding from the Cassidy et al. (2011, 2016) studies that comparative relations play a key role in intellectual potential. Cassidy et al. found significant increases in mean IQ score after training multiple stimulus relations including comparison. Finally, in both Stages 1 and 3 of the present study, the comparison substage was strongly correlated with all substages in Stage 4. These data suggest that nonarbitrary and arbitrary comparative relational repertoires are important intellectual skills and may be predictive of analogical ability. Future researchers may want to further investigate the effects of training nonarbitrary and arbitrary comparative relations in students both with and without language delays.

The present cross-sectional data also showed that scores were lowest for temporal relations across all age cohorts, suggesting that the temporal relation is acquired after the other tested frames. These findings are consistent with cognitive research which has found that temporal understanding emerges relatively late in development (McCormack & Hoerl, 2017; Pyykkönen & Järvikivi, 2012). A noted advantage of the RFT approach adopted in the present study, however, is the analysis of temporal understanding in terms of functional units. The fact that we can conceptualize temporal understanding in terms of relational frames means that we can target this repertoire for training and assess for potential improvements on this basis. Within the RFT literature, temporal framing has previously received less empirical attention than other relations (Hughes & Barnes-Holmes, 2016). It shares the same basic pattern as comparative relations in that it entails responding to events in terms of their directional displacement along a specified dimension. However, time is inherently more abstract than other comparatives, such as size (Hayes et al., 2001b), and requires discriminating successive changes in time. Thus, conceptualizing the physical dimension along which temporal comparatives are arranged requires a more complex verbal repertoire and metaphorical understanding. From an RFT point of view, this might explain why temporal framing appears to be acquired later than other frames. Given the importance of this repertoire, however, further research in testing and training of both nonarbitrary and arbitrary temporal relations is warranted.

In the present study, with the exception of opposite relational responding, participants performed better on the nonarbitrary stages than the arbitrary stages, thus supporting RFT and previous research (Berens & Hayes, 2007; Gorham et al., 2009). In general, it is accepted that nonarbitrary responding is required for the acquisition of arbitrary relational responding (Rehfeldt & Barnes-Holmes, 2009). Previous research has found that participants who did not learn to derive a particular pattern of arbitrary relations were aided in doing so via additional training in the corresponding pattern of nonarbitrary relations (Berens & Hayes, 2007; Gorham et al., 2009). Likewise, in the PEAK assessment and curriculum model, Dixon et al. (2014) train simple and complex verbal relations using nonarbitrary stimuli first, and then progress to equivalence and transformation training.

Given the foregoing, one perhaps surprising outcome of the present study was that participants scored higher in the arbitrary opposition stage than in the nonarbitrary opposition stage. We think it is possible that children scored poorly on nonarbitrary opposition because opposite relations are not typically systematically trained as two dichotomous stimuli on a specified physical dimension on a continuum (Hughes & Barnes-Holmes, 2016). Instead, opposite relations are more likely to be taught as an intraverbal response (e.g., the opposite of tall is short, big/small, full/empty), and thus participants may be more familiar with the arbitrary format (i.e., what is the opposite of *many/few*?) than the nonarbitrary format as in the present study. However, it remains unclear whether participant performance was idiosyncratic to the test or idiosyncratic to their education. In any case, opposite relations are empirically important as shown by Cassidy et al. (2011, 2016), for example, who found that training opposite and comparative relations led to improvement in IQ scores. Future researchers may want to investigate systematically training opposite relations as functional units, as defined by RFT, at both nonarbitrary and arbitrary levels.

Considering the potential for improving language and cognition, further research into relational assessment and training for the purpose of curriculum development is clearly warranted. Up until recently, most empirical work on verbal behavior has largely been influenced by Skinner's (1957) analysis of verbal behavior (Dixon et al., 2017; Dymond et al., 2010). As a result, commonly used language assessments such as the Assessment of Basic Language and Learning Skills-Revised (ABLLS-R; Partington, 2008) and the Verbal Behavior Milestones and Placement Program (VBMAPP; Sundberg, 2008) are based on Skinner's analysis of verbal behavior; thus, training focuses on the basic verbal operants (i.e., echoics, mands, tacts, intraverbals) with little attention to more complex verbal behavior. Dixon et al. (2018) provided experimental work assessing and training more complex language. The present study contributes to the work on relational language assessment.

Previous relational training studies (e.g., Barnes-Holmes, Barnes-Holmes, & Smeets, 2004; Belisle et al., 2016; Berens & Hayes, 2007) showed that derived relational responding can be brought under operant control. A functional analysis of young or developmentally delayed children's existing relational abilities would provide the framework for a robust, flexible, and individualized RFT-based curriculum. The present study is one of the first studies to look at the sequence of acquisition of multiple frames in young children; these findings could be a valuable reference in curriculum design.

Related to this point, one possible limitation in the current study is the representativeness of the sample population. All participants recruited for this study attended the same private school in New Jersey. As mentioned, the average IQ score for participants in this study without the outlier was 112 (113 with the outlier participant), which is slightly above the average range of 90–109 (Roid, 2003). Future research using the present protocol to assess relational framing in young children should include substantially larger numbers both overall as well as within each of the age cohorts. It should also recruit from a wider variety of educational institutions such as public schools and disadvantaged schools. Such extensions of the present work would arguably constitute more representative testing and provide more generalizable results.

Despite this point, the present data are informative in terms of the development of a possible RFT-based curriculum. For example, the present data show correlations between age and both nonarbitrary and arbitrary relational ability. These normative data could help inform a *developmentally sequenced* relational curriculum; for example, a logical training sequence would include training increasingly difficult or complex levels of nonarbitrary relational frames before training arbitrary relational frames. Another area of warranted research includes testing and training the transition from nonarbitrary to arbitrary relations; a comprehensive relational curriculum should include training-for-transition procedures.

Additional within-stage analyses show some interesting results with respect to particular relational frames which could be considered when designing relational programs. For example, in Stage 1, there was a significant correlation between comparison and age, IQ, total relational assessment scores, and total substage scores for each substage. Future researchers could investigate training nonarbitrary comparative relations in order to facilitate the development of other frames. In Stage 2, nonarbitrary analogical frames of coordination and hierarchy correlated with age; therefore, it may be worth training these frames first in nonarbitrary analogy programs. In Stages 3 and 4, all substages correlated with age, and in particular, the data showed at which age particular frames were acquired. In Stage 3, the biggest age-based improvement in arbitrary relational scores occurred between the 3- to 4-year-old and 4- to 5-year-old cohorts. These data are consistent with the Mulhern et al. (2017) study, which found that 3-to 4-year-old participants did not respond correctly on tests of hierarchical framing, whereas participants aged 5 and older performed better on these tests. Based on these data, considerable development in arbitrary relational framing occurs between ages 4 and 5. In addition, the data from Stage 3 in the present study indicate that coordination is acquired first, and temporality is acquired last. Thus, a relational language program could introduce relational training around the developmental age of 4, and start with training the relational frame of coordination first. The previously mentioned VB-MAPP (Sundberg, 2008) is based on the verbal repertoires of typically developing children up to 4 years old; thus, the development of a robust relational assessment and curriculum suitable for the developmental age of 4+ is warranted.

Analogy

Apart from examining the acquisition of relational framing in general with age, the present study also focused considerable attention on relating relations, that is, analogy. The study assessed analogy in young children at both nonarbitrary and arbitrary relational levels. It is commonly accepted in mainstream cognitive research (e.g., Bod, 2009; Gentner, 1983) that analogy is an important pattern of relational framing with regard to intellectual potential. Moreover, analogical reasoning is frequently applied as a metric of intelligence (Sternberg, 1977), and as a measure to predict academic success; for example, the Law School Admissions Test (LSAT; Lapiana, 2004). The theoretical importance of analogy has also been recognized explicitly within the RFT literature. For example, the relating of relations is included as one of the five levels (including mutual entailing, relational framing, relational networking, relating relations, and relating relational networks) in Barnes-Holmes et al.'s (2017, 2020) recently proposed hyperlevel multidimensional (HDML) framework for examining and discussing relational framing. However, despite the ubiquity and utility of analogy, functional assessments targeting relations among relations as analogy have not yet been examined.

The present data extend previous work on assessment of analogy by Carpentier et al. (2002, 2003, 2004). However, the present study arguably advances the Carpentier et al. studies by using a more time-efficient format than the match-to-sample (MTS) procedure used by Carpentier et al. (this will be discussed at greater length further ahead). In the present study, two of the stages (Stages 2 and 4), addressed the relating of relations; Stage 4 demonstrated the RFT definition of analogy, or deriving relations between relations, whereas Stage 2 tested the nonarbitrary precursor to analogical responding.

In Stage 4, the total score as well as individual substage scores show significant correlations with age, IQ score, and total assessment scores. Total score for arbitrary analogical responding showed a slightly stronger correlation with IQ performance than basic arbitrary relations (Stage 3), whereas basic arbitrary relations showed a slightly higher correlation with age compared to arbitrary analogical relations. These data provide further evidence that analogical relations are undoubtedly tied with intellectual potential.

The results for Stage 4 showed a marked difference in scores between the 4-to 5-year-old cohort and the 5-to 6-year-old cohort. These findings contribute to the extant RFT research suggesting a developmental divide in the acquisition of analogical ability at around 5 years of age (Carpentier et al., 2002, 2003). The 5-year-old participants in the Carpentier et al. studies were able to demonstrate arbitrary equivalence–equivalence and nonequivalence–nonequivalence only after additional compound-matching training with the trained relations between elements (e.g., A1B1-A3B3). The mean scores for the 6- to 7-year-old cohort were better again and were more evenly distributed across frames. In light of the strong correlation between basic arbitrary relations and age mentioned above, it is possible that an additional year of practicing deriving relations (Stage 3) facilitated relating relations within relations in Stage 4.

Unlike the Carpentier et al. studies, however, the present study examined both nonarbitrary (Stage 2) and arbitrary (Stage 4) analogical responding across five frames, providing further insight into the developmental sequence and acquisition of analogical responding. Total relational assessment scores for Stage 2 (nonarbitrary analogical relations) showed strong, significant correlations with age, IQ score, and total assessment scores. Data were further analyzed for patterns of responding by age group in order to identify at which age analogical responding is acquired. Stage 2 results showed a gradual improvement in scores by age, and suggested the acquisition of analogy, at least in coordinate and difference relations, at 5 years of age. This is the first RFT study to focus on nonarbitrary analogy in addition to arbitrary analogy. In the domain of comparative psychology more work has been done testing relating of nonarbitrary relations in nonhumans; indeed, various species have been found to pass such tests. For example, chimpanzees (Gillan et al., 1981), crows (Smirnova et al., 2015), and baboons (Fagot & Maugard, 2013) have passed nonarbitrary analogical (i.e., relating relations) tasks. In addition, such work has also been conducted by cognitive-developmental researchers (e.g., Christie & Gentner, 2014). The present study begins to bridge that gap in the RFT-based study of nonarbitrary analogy in young children. From an RFT perspective of course, nonarbitrary analogy is not *full* analogy (i.e., deriving relations among arbitrary relations). However, it is an important repertoire and should be trained as a prerequisite skill before arbitrary analogy. The strong correlations between nonarbitrary and arbitrary analogy total scores in the present study provide evidence for this analysis, and future researchers could further evaluate the effects of training nonarbitrary analogy if it is a weak or missing skill in young children.

Finally, an additional point that might be made regarding the importance of analogy concerns the data from Participant S26, an outlier in the 3- to 4-year-old cohort. Participant S26's full-scale IQ score was 132; without his data, the mean IQ score for all participants was 112. It is interesting that Participant S26's data are similar to those of his cohort for the nonarbitrary relations and nonarbitrary analogy stages. However, his scores were above the mean in arbitrary relations and arbitrary analogy; he was the only participant under 4 years of age to respond correctly to arbitrary analogy tasks, thus positively skewing the results of the 3- to 4-year-old cohort. He also outperformed the 4-to 5-year-old participants in the arbitrary analogy stage. His data provide further evidence that arbitrary analogical responding is strongly tied with intellectual potential.

Relational Assessment Format

One potentially important aspect of the present study was the assessment format. Previous RFT studies have used a number of different methodologies to examine and train relational framing in young children, including, for example, the MTS (see Rehfeldt & Barnes-Holmes, 2009) procedure and the relational evaluation procedure (REP; see Barnes-Holmes et al., 2001; Stewart et al., 2004). MTS procedures require extensive baseline training prior to starting any testing or training. For example, in the Carpentier et al. (2002) study, the 5-year-old participants required an average of 234 baseline trials in Experiment 1 before training and testing could begin. The REP, on the other hand, is a more efficient method in which multiple exemplars of relational patterns can be easily generated and readily presented without requiring any prerequisite MTS training. This methodology allows participants to report on or evaluate sets of arbitrarily applicable relations defined by various sets of contextual cues. The testing format utilized in the current study was employed in the context of the multistage relational assessment that allowed testing of a range of different types of relations across different levels of complexity using the REP. The relational networks in the present study were displayed on the screen during each trial in order for participants to derive relations and relate relations among relations in the arbitrary stages, thus omitting the need for time-consuming MTS pretraining.

In order to make the present relational assessment equally accessible to all participants, including younger cohorts unable to read, a novel procedure was implemented, which excluded any textual instructions or tasks. Instead, this measure used simple, monochromic shapes and single, alphabetic letters to indicate the contextual cue (i.e., *S/D, M/L, S/O, B/A,* and *C/I* for same/different, more/less, opposite, before/after, and contains/inside, respectively). In addition, the letter identifying the contextual cue was paired with optional audio stimuli, for example, "same" for *S*.

A functional analysis of intelligence as relational skills, such as the analysis provided by the relational assessment in this study, provides useful information on the relational skills that contribute to intelligent behavior as assessed on IQ tests. Considering the predictive validity of IQ scores (Sternberg et al., 2001), understanding the variables that influence intelligent behavior is a socially valid endeavor. Furthermore, the extant RFT research on relational training and intellectual performance demonstrates that relational framing can be trained and strengthened (e.g., Barnes-Holmes, Barnes-Holmes, & Smeets, 2004; Barnes-Holmes, Barnes-Holmes, Smeets, Strand, & Friman, 2004; Belisle et al., 2016; Berens & Hayes, 2007). Thus, identifying weaknesses in relational ability could provide the scaffolding needed for a relational curriculum (discussed in more detail below). The relational assessment protocol used in the present study can provide a useful instrument in such work going forward.

Despite its advantages, the relational assessment measure used herein may also need some further refinement. Analyses of the present data reveal certain trends that are interesting; nevertheless, we have to bear in mind that the format of the assessment may have made some relations more difficult than they would be if we had given participants a symmetry test or a MTS task, for example. In addition, because the assessment is constrained in what it tests in each particular frame, we can make some comparisons between individual frames, but it is hard to be absolute or definitive about relational ability. For example, 1) we can clearly see that across the four stages, coordination is a simpler relation than the other relations; 2) in Stage 3, we can visibly see an improvement in substages across age; and 3) in Stage 4, we can see the acquisition of analogical responding at age 5, but other comparisons are not as conclusive. However, the primary purpose of the assessment is to provide a general overview of a child's nonarbitrary and arbitrary relational repertoire. In order get a more comprehensive measure of a child's relational abilities, future researchers could develop a relational protocol that systematically assesses direct relations, entailed relations, and transformation of stimulus function with more scope and depth. For example, this could be done by adding more detailed tasks to each substage, one stage at a time, or by focusing on the evolution of nonarbitrary to arbitrary relational framing across age groups, one frame at a time. Furthermore, additional relations such as distinction and deixis could be included in future relational assessments. Approaching relational training in such a functionally specified way will enable language protocols to assess and train more generative, flexible, and complex language.

A second potential limitation of the relational assessment may be the omission of distinction as an independent frame. In the present study, distinction was not tested in Stage 1; and in Stages 2–4 distinction was included in the coordination substage in order to provide a comparison option. More work on distinction is warranted and future researchers may consider testing coordination and distinction more systematically and providing independent analyses for both frames.

A third potential limitation of the relational assessment may be the *linear* versus *nonlinear*, and *same* versus *mixed* presentation of the arbitrary relational networks in Stages 3 and 4. In a linear-same series, stimuli are presented sequentially (e.g., A–B–C) and the contextual cue between relata is the same. For example, in a linear-same comparative relational network, A is more than B, and B is more than C, or A > B, B > C (Fienup & Brodsky, 2020; Vitale et al., 2008). In a linear-mixed series, stimuli are presented sequentially but the contextual cue is not the same between relata, for example, A > B, B < C. In nonlinear series, stimuli are not presented linearly (e.g., A–B, C–A); in a nonlinear-same series the contextual cues are the same between relata, and in a nonlinear-mixed series the cues are not the same. For example, A > B, C > A and B > A, C < A, respectively (Vitale et al., 2008). In Stage 3 in the present study, the coordination substage included linear-same, linear-mixed, and nonlinear-same relational networks, and all other substages (comparison, opposition, temporality, and hierarchy) included only linear-same relational networks. In order to distinguish between sameness and difference, the difference relation had to be included in the coordination relational networks, thus resulting in mixed networks. For all substages including coordination, however, questions regarding relations in the network always progressed from directly trained, to mutually entailed, to combinatorially entailed relations.

In Stage 4, the coordination and opposition substages included linear-mixed networks, and the comparative, temporal, and hierarchical substages included nonlinear-mixed networks. Deriving relations from nonlinear networks is more difficult than deriving from linear networks (Hunter, 1957; Vitale et al., 2008). However, all substages included mixed contextual cues, and as in Stage 3, trials progressed from directly trained, to mutually entailed, to combinatorially entailed analogical relations. Future replications of Stages 3 and 4 should include systematic transitions from linear-same-mixed to nonlinear-same-mixed relational networks in order to gain more detailed information regarding participants' relational skills.

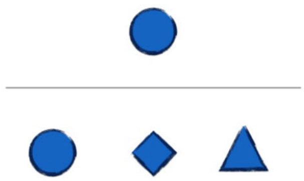
Conclusion

Research in RFT has produced considerable proof that relational framing is a core skill underlying language and cognition. However, despite the evidence, research into the assessment and training of relational responding remains limited. The present data contribute to the exiguous literature on relational testing and training, and initiate the RFT-research on the normative development of frames. The results from the present study could be referred to as a starting point for further research in the normative development and sequence of relational frames, and for relational skills assessment and training.

Appendix A1

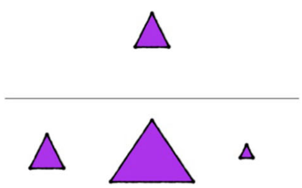
Relational Assessment Stage 1: Nonarbitrary Relations

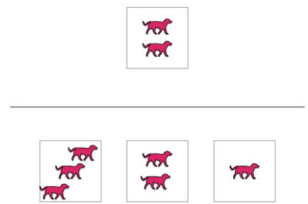
Coordination



Example: Which one is like the one at the top (instructor points to sample)?

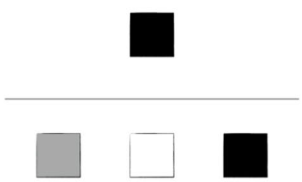
Comparison

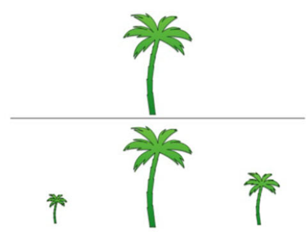




Example: Which one is bigger/smaller than the one at the top (instructor points to sample)?

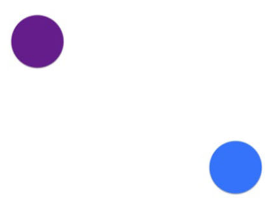
Opposition





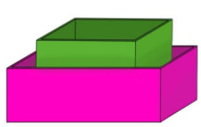
Example: Which one is opposite to the one at the top (instructor points to sample)?

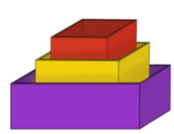
Temporality



Example: Which x is before/after the other y? Was the x before/after the y?

Hierarchy



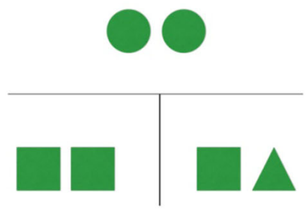
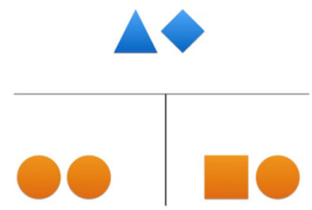


Example: Which one is inside? Which one contains the other one? Does x contain y? Is x inside y?

Appendix A2

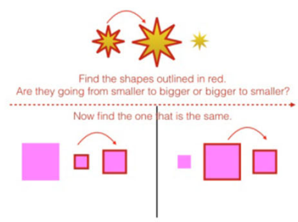
Relational Assessment Stage 2: Nonarbitrary Analogical Relations

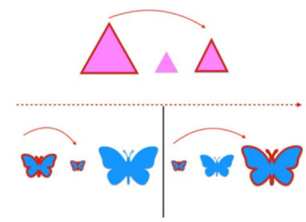
Coordination

Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

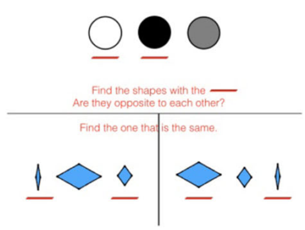
Comparison

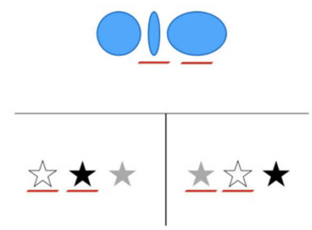




Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

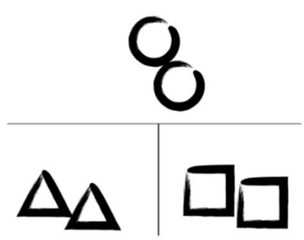
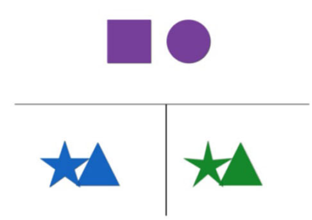
Opposition





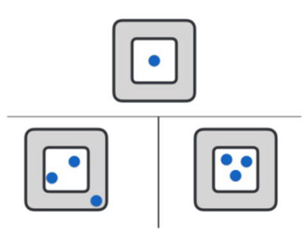
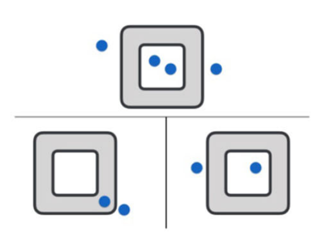
Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

Temporality

Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

Hierarchy

Example: Which one of these (points to comparisons) is like the one at the top (points to sample)

Appendix A3

Relational Assessment Stage 3: Arbitrary Relations

Coordination

Example: Does x like the same food as y? Does x like different food than y?

Comparison

Example: Which coin would you take to the store?

Opposition

Example: Which coin would you take to the store?

Temporality

Example: Is x before/after y?

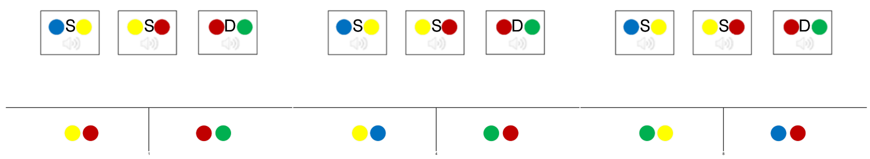
Hierarchy

Example: Is x inside y? Does x contain y?

Appendix A4

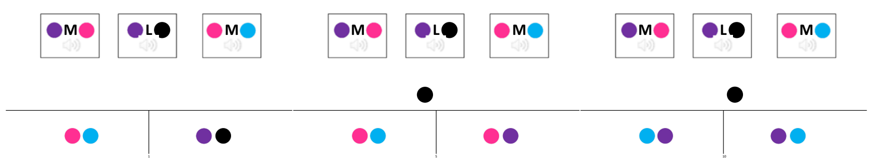
Relational Assessment Stage 4: Arbitrary Analogical Relations

Coordination



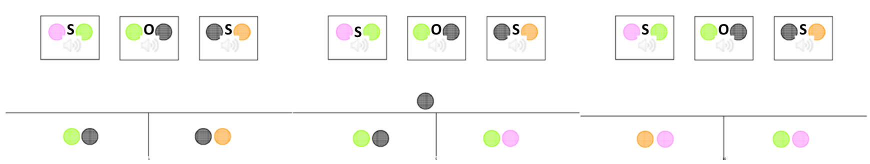
Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

Comparison



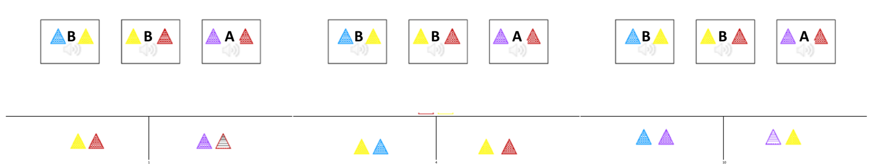
Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

Opposition



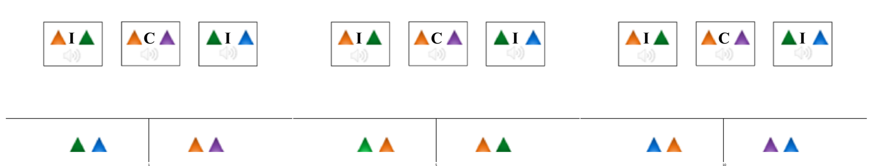
Example: Which one of these (points to comparisons) is like the one at the top (points to sample)

Temporality



Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

Hierarchy



Example: Which one of these (points to comparisons) is like the one at the top (points to sample)?

Appendix 2

Table 4 Chance level responding across stages and substages

	Stage 1	Stage 2	Stage 3	Stage 4
Coord	33%	50%	38%	50%
Comp	33%	50%	40%	50%
Opp	33%	50%	40%	50%
Temp	50%	50%	42%	50%
Hier	45%	50%	40%	50%

This table represents chance level responding, or what you might expect a child to get correct purely on the basis of chance or consistently guessing. These data can be used to compare with participant relational assessment scores

Availability of data and material The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Code availability Not applicable

Funding Not applicable

Declaration

Conflicts of interest The authors declare that they have no conflict of interest.

Ethical Approval The study was approved by the NUI Galway Research Ethics Committee. All procedures performed in the studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent to participate Informed consent was obtained from all individual participants included in the study.

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